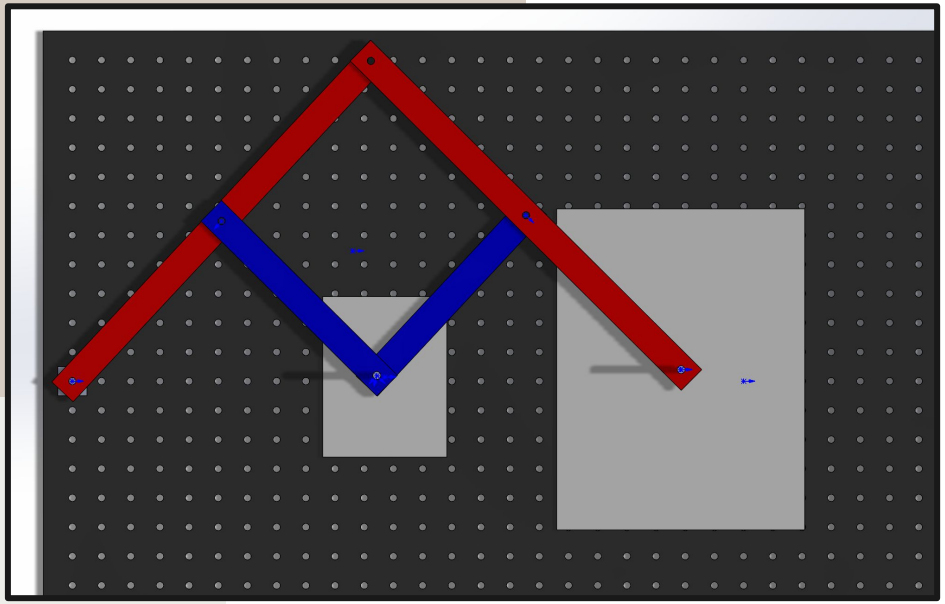
Design and Manufacturing: Pantograph

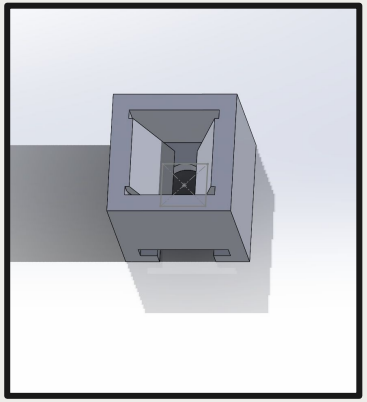


By: Kasin Ito-Sato and Evan Soleimani

Parts List and Machining Method

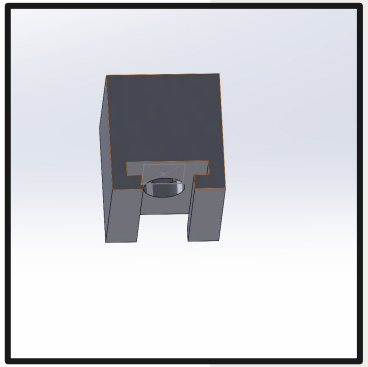
- Stainless Steel Bolts ¼”-20 (QT: 4)
- Nuts ¼”-20 (QT: 4)
- Aluminum Bars (QT: 47”)
  - Sawing and Milling
- 3D Printer Polymer
  - 3D Printing



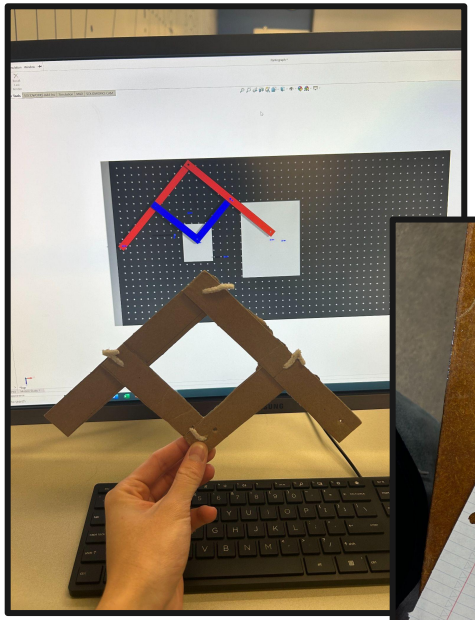


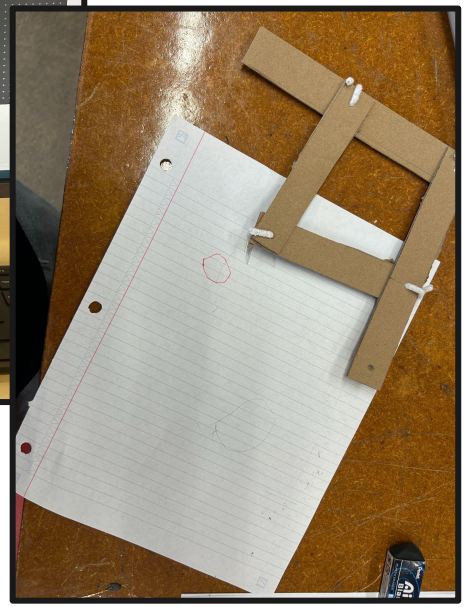
Potential:

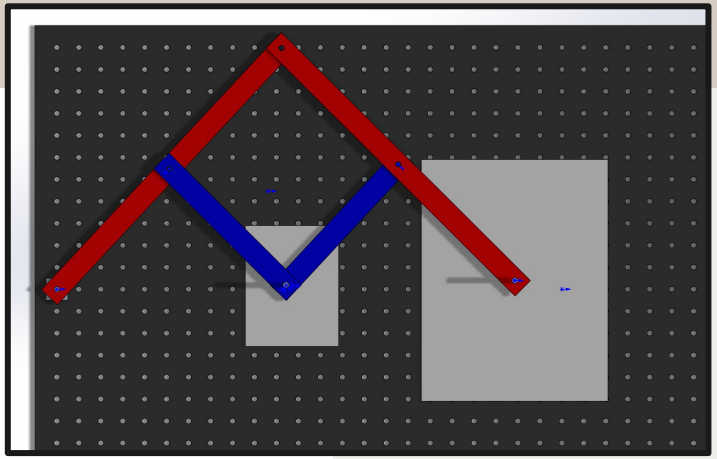
- Acrylic sheets (cut four pieces out of it)
  - Laser Cutter

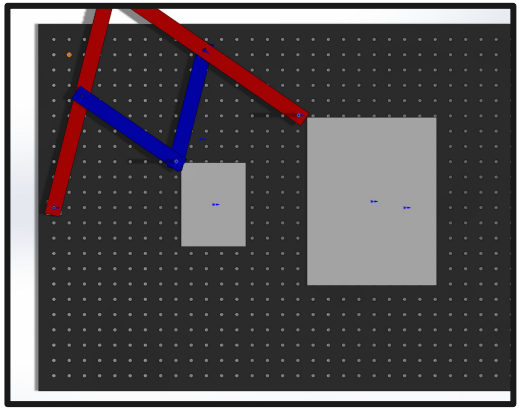


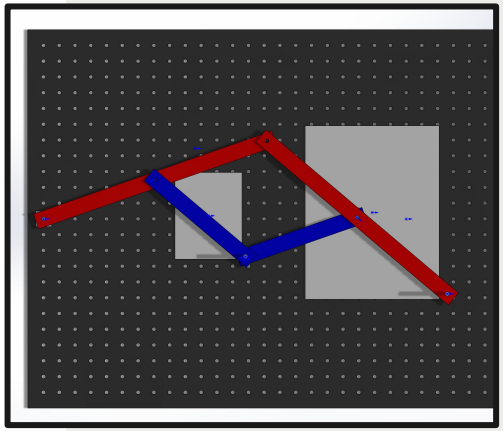
Pantograph and “Soft” Prototype











Tolerance Analysis

- Stainless Steel Bolts ¼”-20 (QT: 4)
  - 2B bolts, 0.25in
  - Fastener MMC: 0.25in
- Hole diameter 0.25in (+0.01in)
  - Hole MMC: 0.26in
- T=H-F
  - T=0.26-0.25=0.01in
  - Floating so half and half
    - 0.01in and 0.01in

